

Case 2

02582 Computational Data Analysis

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Introduction/Data description

The aim of this project is to investigate whether low-dimensional representations of wearable biosignal features reveal experimental and emotional states in the EmoPairCompete dataset. The central research question is:

Can representations of HR, EDA and TEMP features separate resting, puzzle/stress and recovery phases, or are these representations mainly dominated by individual physiological baselines?

The dataset contains 312 observations from 26 individuals, corresponding to 4 rounds and 3 phases per individual. The extracted feature file contains 51 physiological variables from HR, temperature and EDA signals, 6 metadata/grouping variables and 11 questionnaire variables. The three phases are pre-puzzle/rest, puzzle/stress and post-puzzle/recovery. The questionnaire variables include frustration and ten PANAS-style emotions, split into positive emotions (active, alert, attentive, determined, inspired) and negative emotions (afraid, ashamed, hostile, nervous, upset).

Table 1: Dataset overview used in the analysis.

Quantity	Value
Observations	312
Individuals	26
Rounds	4
Phases	3
Physiological features	51
Emotion/questionnaire variables	11
Total missing entries	9

Model and method

The analysis follows the required pipeline: data pre-processing, representation and evaluation. First, physiological input variables were separated from metadata and questionnaire variables. The metadata variables were not used as model inputs, but were used for coloring, grouping and validation. Missing numerical values were handled by median imputation. All physiological features were standardized before PCA, clustering, classification diagnostics and CCA. Scaling is necessary because the variables have different units and magnitudes, for example HR, temperature and EDA-derived quantities.

The main representation method was principal component analysis (PCA), used as a transparent linear baseline. PCA is useful here because it summarizes correlated physiological features into a few orthogonal directions and provides interpretable feature loadings. To check whether the representation aligns with the experimental design, the first PCA scores were visualized by phase and by

individual. A K-means clustering check with $K = 3$ was also applied to the first five PCA components and compared with the three experimental phases using adjusted Rand index (ARI). This clustering step was only used as an unsupervised diagnostic, not as the main result.

As a supplementary analysis, canonical correlation analysis (CCA) was used to examine whether combinations of physiological features relate to combinations of emotion ratings. CCA was run separately for positive emotions, negative emotions and all questionnaire variables. The CCA results were interpreted cautiously, because CCA is supervised with respect to the questionnaire variables and can overstate structure if validation is not subject-aware.

Model selection

For PCA, the number of components was selected by explained variance and by keeping the representation simple. The first component explained 27.8% of the physiological variance, the second explained 10.0%, and the first five components explained 58.4%. Therefore, the first two components were used for visualization and the first five components were used for the quantitative phase diagnostic.

For K-means, $K = 3$ was fixed before fitting because the experimental design has three known phases. This was not optimized from the labels. For CCA, one component was used in validation to avoid an overly flexible model and because the first canonical direction is the easiest to interpret. The final CCA fitted on all observations was used only for interpretation of loadings, while validation results were reported separately.

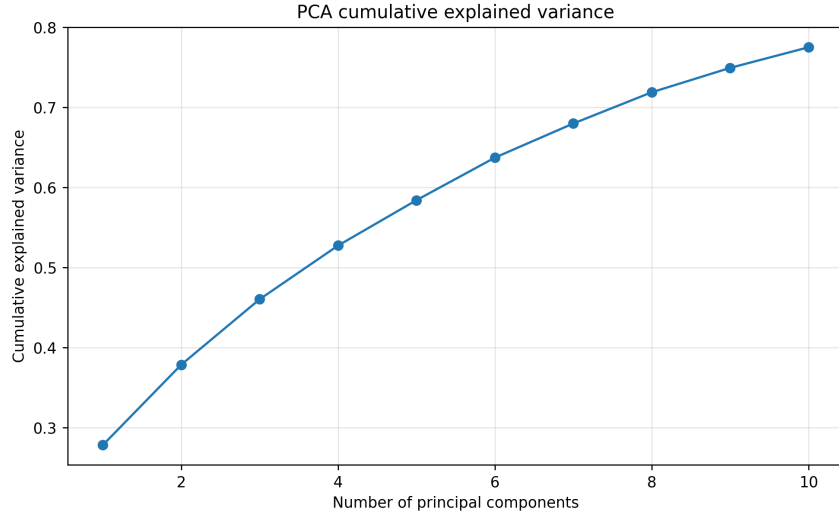


Figure 1: Cumulative explained variance for PCA on standardized physiological features.

Missing values

The dataset was nearly complete. Only 9 individual entries were missing across the full feature table. Missing values appeared in EDA recovery/rise-time variables and a few emotion ratings. Since the missingness was small compared with 312 observations, median imputation was used for numerical variables inside the analysis pipeline. This keeps the sample size unchanged while avoiding leakage, because imputation is fitted inside each training fold during cross-validation. Dropping rows would also have been reasonable, but it would remove complete physiological recordings due to a very small number of missing entries.

Factor handling

Categorical variables were handled as factors for interpretation and validation, not as physiological inputs. The variable **Individual** was used as the grouping variable in subject-aware validation. **Phase** was used as the experimental condition for plot coloring and evaluation. **Round**, **Puzzler** and **Cohort** were kept as descriptive factors. They were not included directly in PCA or CCA input matrices, because the main question concerns representations derived from biosignals rather than from metadata.

To understand how much questionnaire variation is associated with factors, eta-squared values were computed between categorical variables and emotion ratings. This showed that individual identity explains a non-negligible part of several emotion ratings, supporting the need for subject-aware validation.

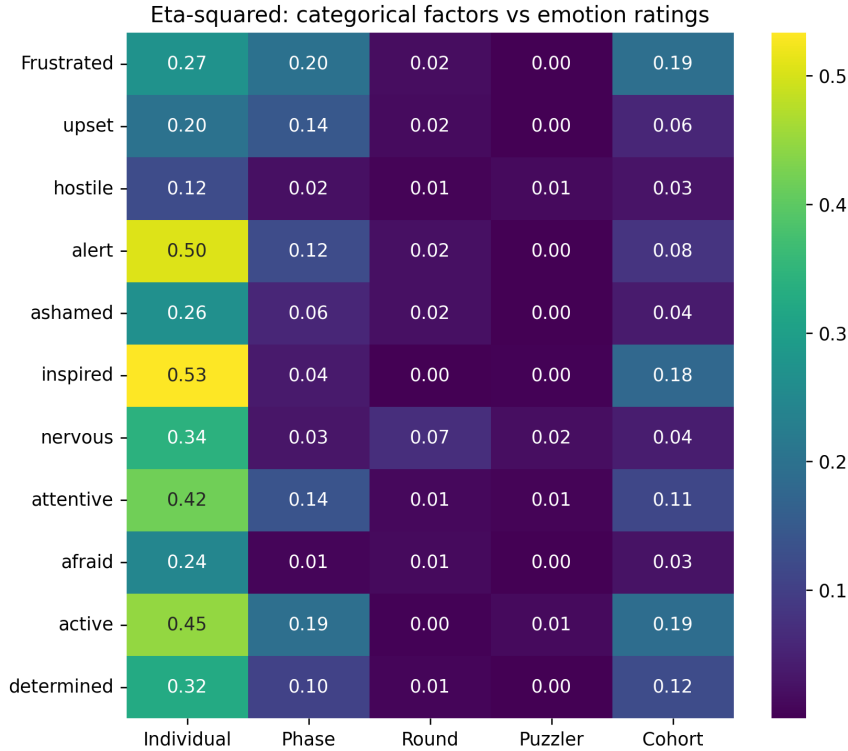


Figure 2: Eta-squared values between categorical factors and questionnaire variables. Individual effects are visible, especially for some positive-emotion variables.

Model validation

The key validation issue is that the dataset contains repeated measurements from the same individuals. A random split can place recordings from the same participant in both training and test folds, making performance too optimistic because the model may learn individual baselines rather than general physiological states. For this reason, subject-aware GroupKFold validation was used with **Individual** as the grouping variable.

Two validation checks were used. First, a diagnostic phase classifier was trained using median imputation, scaling, five PCA components and balanced logistic regression. This classifier was not the main model; it was used to quantify whether the learned representation carries phase information. Random stratified 5-fold validation achieved 0.459 balanced accuracy, while subject-aware GroupKFold achieved 0.441 balanced accuracy. Both values are only moderately above chance for three balanced classes, indicating weak phase separability.

Second, CCA was evaluated by measuring the correlation between held-out physiological and emotion canonical scores. Positive emotions had mean canonical correlation 0.243 under random 5-

fold validation, but this decreased to 0.121 under subject-aware GroupKFold. Negative emotions had approximately no subject-aware canonical correlation (-0.007). The all-questionnaire CCA setting reached a subject-aware correlation of 0.218, but this value should still be treated as exploratory because fold-level scores were variable. This drop from random to subject-aware validation is important: it suggests that some apparent physiology-emotion association may be partly subject-specific and does not strongly generalize to unseen individuals.

Table 2: Validation summary. Higher values are better, but the subject-aware results are the main estimate.

Analysis	Random CV	Subject-aware GroupKFold
Phase diagnostic, balanced accuracy	0.459	0.441
CCA positive emotions, canonical correlation	0.243	0.121
CCA negative emotions, canonical correlation	0.053	-0.007
CCA all questionnaire variables, canonical correlation	0.288	0.218

Results

The PCA plots show that the first two components summarize a substantial part of the physiological variation, but they do not cleanly separate the three phases. The phase-colored plot shows overlapping clouds, while the individual-colored plot suggests that repeated measurements from the same individuals contribute strongly to the geometry of the representation. This supports the interpretation that individual baselines are an important source of variation.

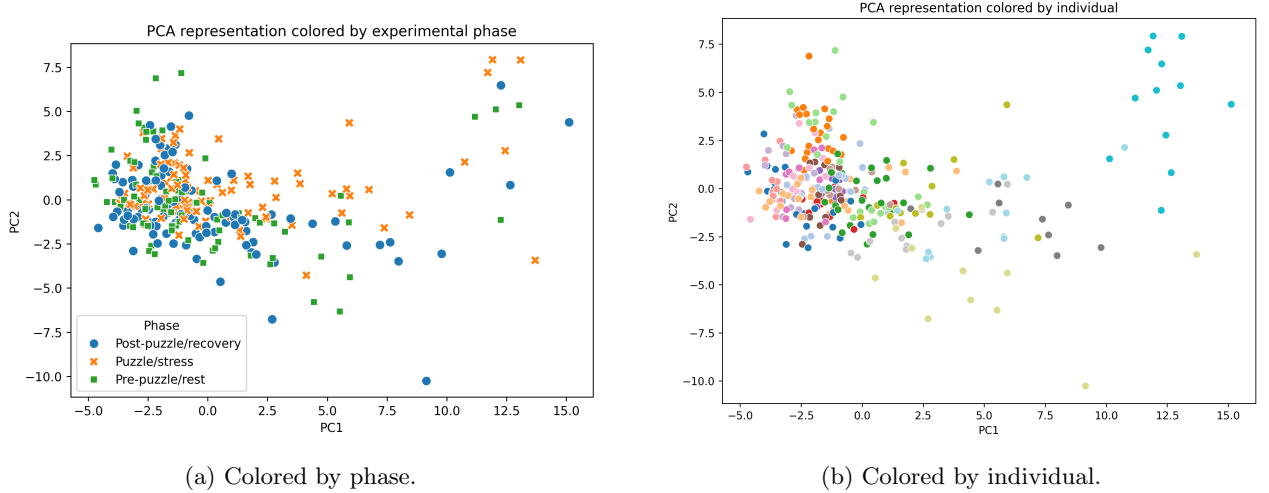
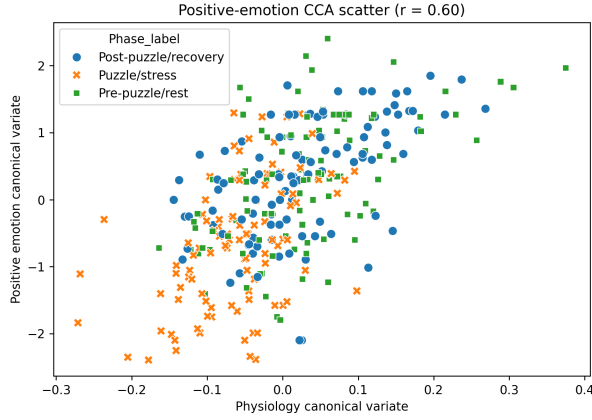


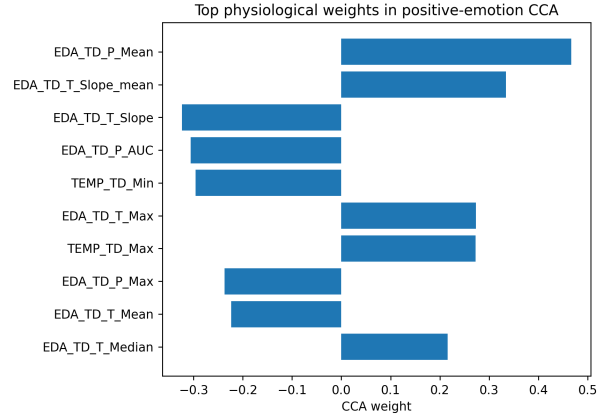
Figure 3: PCA representation of physiological features. Phase separation is weak, while individual-specific variation is visible.

The clustering result supports the same conclusion. K-means clustering on the first five PCA components had $ARI = -0.002$ against the true phase labels, meaning that unsupervised clusters did not match the experimental phases. This does not mean the analysis failed; rather, it indicates that the dominant unsupervised variation in the physiological features is not simply the three-phase experimental structure.

CCA gave a clearer relation when fitted on all observations: the first positive-emotion CCA component had correlation $r = 0.603$. The most influential physiological weights were EDA phasic mean, EDA tonic slope variables, EDA phasic AUC and minimum temperature. These features are plausible because EDA dynamics and temperature can reflect arousal and physiological regulation. However, the subject-aware validation correlation for positive emotions was only 0.121, so the full-data CCA should be interpreted as an exploratory association, not as a robust predictive model for new individuals.



(a) CCA scores fitted on all observations.



(b) Largest physiological CCA weights.

Figure 4: Positive-emotion CCA interpretation. The full-data association is visible, but subject-aware validation shows weak generalization.

Overall, the project finds weak evidence that low-dimensional physiological representations reliably separate experimental phases across individuals. There is a moderate exploratory relationship between physiology and positive emotion ratings, but this relationship becomes much weaker under subject-aware validation. The main conclusion is therefore that the dataset is strongly affected by noise and individual baselines. A rigorous pipeline should not rely on random splits alone, because random validation can overestimate the generalizability of biosignal representations.

Limitations and conclusion

The main limitation is the small number of individuals relative to the amount of between-subject variability. Each person contributes repeated recordings, so subject-specific baselines are difficult to separate from experimental effects. The analysis also uses pre-extracted summary features rather than full time-series dynamics. Future work could use richer temporal features, subject-level baseline normalization or a simple autoencoder representation followed by the same subject-aware evaluation.

The final conclusion is that PCA and CCA provide useful exploratory representations, but the phase and emotion signals are weak when evaluated on unseen individuals. This is a scientifically meaningful result for wearable biosignals: the data should be interpreted with caution, and subject-aware validation is necessary before claiming that physiological features reveal stress or emotion states reliably.

Acknowledgement of GenAI use

Generative AI was used as a support tool for discussing the project structure, drafting and revising code organization, debugging local execution steps, and improving the wording of the report. Suggestions were reviewed and adapted before use. I take responsibility for the submitted report, code, figures, and conclusions.

References

- [1] S. Das, N. L. Lund, C. R. Gonzalez, N. N. Lonfeldt, and L. H. Clemmensen. EmoPairCompete - physiological signals dataset for emotion and frustration assessment under team and competitive behaviors. ICLR Workshop on Learning from Time Series for Health, 2024.
- [2] E. R. Thompson. Development and validation of an internationally reliable short-form of the positive and negative affect schedule. Journal of Cross-Cultural Psychology, 2007.